

Chapter One

Adaptive Web Interfaces Using Constrained Reinforcement Learning and Human-in-the-loop

1.0 INTRODUCTION

The introduction of the World Wide Web has transformed contemporary society as the main source of information exchange, trade, and communication.

Nevertheless, the existing design paradigm remains largely unchanged even though web technologies have evolved at a high pace. A majority of web interfaces in the current market are constructed as being one-size-fits-all, that is to accommodate the large base of average users. While this strategy may be effective for development, it inevitably affects the user experience (UX) of a certain segment of the population, especially individuals with unique needs, different interests, or physical impairments. This fixed model is increasingly regarded as obsolete in the context of modern information technology (IT) solution where personalization and inclusiveness become central. Artificial Intelligence (AI) and Reinforcement Learning (RL) provide a strong answer to this dilemma. The Reinforcement Learning agents are capable of learning the best policies to be used in changing User Interface (UI) contents (e.g., layout, font size, and navigation structure) depending upon the real-time user interaction. Having a user interface as an environment and user satisfaction as a reward signal, an AI agent can be dynamically configured in order to adapt a website to the specific context of the visitor. Nonetheless, the introduction of AI to web design presents a lot of risks. An unconstrained Reinforcement Learning Agent that maximizes its specified reward function (e.g., time on site or click-through rate). A naive agent can unintentionally create interface settings that do not meet the requirements of accessibility, which include the Web Content Accessibility Guidelines (WCAG). To give a scenario, an agent may decrease the contrast color so that a design appears to look sleek, which may lead the design to look unreadable to people with poor vision. Moreover, the self-generated modifications will confuse the users, making them lose control and faith in the system. This current project suggests a new structure: “Adaptive Web Interfaces through Constrained Reinforcement Learning and Human-in-the-loop”. This method has the advantage of utilizing the optimization ability of

Reinforcement Learning, but within safety limits set by WCAG guidelines. It will further overlay a Human-in-the-Loop (HITL) to check the adaptations to make sure the system is trusted and user-centered. This study is aimed at leading to the provision of a solid IT solutions in order to increase the need for an inclusive and intelligent web by resolving the two-fold problem of accessibility and user trust.

1.1 BACKGROUND OF THE STUDY

The journey of the web interface design has been one of increasing abstraction and adaptability. In the creation of the web in the 1990s, the pages written in Hypertext Markup Language were primarily static pages. This meant that the styling of the page was directly connected with the content, and the functionality of the page was strictly limited to the capabilities of the web browsers of that time. The evolution of Cascading Style Sheets (CSS) in the late 1990s brought the first big change in web development. With the rise of mobile devices in the late 2000s came a crisis in the industry, desktop oriented websites were very unstable for mobile devices. This would lead to the popularity of Responsive Web Design, a term coined in 2010 by Ethan Marcotte. Responsive Web Design employed the concept of “media queries,” which were used to respond to viewing device size and changing of layouts. Responsive Web Design was a huge improvement, but still a reactive technology that depended on the device attributes rather than those of the user. A 60-year-old user with low vision and a 20-year-old user with perfect vision would receive the same interface if they were using the same device, despite having different needs. In recent years, the focus has been shifted towards “Adaptive” interfaces that can respond to user data. Early attempts involved rule-based systems, which are designed so that if a user is on a slow connection, load low-resolution images. However, these rules were manually hard coded by developers and could not scale to the complexity of human behaviour. The advent of Modern Machine Learning brought about the next evolution: an Artificial Intelligence driven personalization. Recommender Systems (like those used by Netflix or Amazon) began adapting content dynamically. Of course, a similar line of reasoning began to be applied to the interaction itself, too. A promising candidate for meeting this requirement was Reinforcement Learning, a subfield of Machine Learning in which an agent learns through trial-and-error learning. Experiments conducted using Contextual Bandits (a simplified variant of Reinforcement Learning) show agents choosing optimal arrangements for maximizing

engaging users. Despite all these breakthroughs, disadvantages still exist. Most current research on Artificial Intelligence and User Interface has only emphasized on so called “performance” factors such as conversion rates. There has been a lack of focus on constraints related to accessibility. Previous projects had mostly thought of accessibility as an addition to User Interface design. This led to optimizing systems that were more active but were inaccessible to those with physical or cognitive disabling conditions. Also, the black box properties of deep models have led to a lack of trust. Users don’t trust interfaces that behave erratically. This project will resolve all these issues by using formal constraints on safety and Human-in-the-loop systems in the adaptive loop.

1.2 STATEMENT OF PROBLEM

Despite continuous advancements in web technologies, many contemporary web interfaces still adopt uniform design approaches that fail to accommodate the diverse abilities, preferences, and accessibility needs of users. Recent studies indicate that while adaptive web interfaces have gained attention as a means of improving user experience, most existing solutions primarily focus on optimizing performance-driven metrics such as engagement, efficiency, and interaction frequency, with limited emphasis on accessibility and inclusiveness (Zhang & Wang, 2023). Many emerging adaptive interface systems employ reinforcement learning and similar approaches to sequence decisions for adjusting interface components based on user behavior. However, the unrestricted model of a learning approach is focused on the maximization of the reward value, which might also lead to the degradation of the user interface for people with disabilities, where the requirement for accessible interface design is not explicitly stated during the adaptation process (Lattimore & Szepesvári, 2020, Trewin, 2018). Recent studies on interface accessibility clearly pointed out the failure of machine learning models in adhering to interface standards, such as Web Content Accessibility Guidelines, which might create interface difficulties for people with disabilities (Krstić et al., 2025, W3C, 2018). Besides issues of accessibility, it must be noted that the autonomous and opaque nature of adaptive artificial intelligence systems provides issues concerning issues of transparency and trust as well as issues of fairness to the end users involved with such systems. Users may become uncomfortable with changes in an adaptive interface as it may be considered unfair due to the lack of human oversight in decision-making

processes involved with adaptive artificial intelligence systems, thus affecting their trust with adaptive systems in general (Lee, 2021). Recent research into human-AI interactions highlights the need to adopt human-in-the-loop components for improved reliability and trust with adaptive systems for greater human-AI collaboration for efficiency and reliability in human-AI interactions that have gained even greater importance with the advent of Web 4.0 (Amershi et al., 2019; Gaspar-Figueiredo et al., 2025).

1.3 AIM AND OBJECTIVES

This study aims to develop an adaptive web interface using constrained reinforcement learning and human-in-the-loop. The Objectives of the study are as follows:

- **Research and Fact-Finding:** To solicit reports and assess extensive literature on the area of Reinforcement Learning in HCI, web accessibility standards (WCAG), and Human-in-the-Loop approaches.
- **System Design:** To develop a suitable conceptual model or framework (the A2-RL Framework) that can successfully superimpose Constrained Reinforcement Learning with a WCAG Safety Shield and a HITL feedback mechanism.
- **Implementation:** To apply the designed framework on a simulation environment (using a representative web application) to prove the capability of the agent to learn safe adaptation policies.
- **Evaluation:** To assess the developed system in terms of selected indicators, in particular, the rates of WCAG compliance, the time spent on each task by the user persona, and the rates of adaptation acceptance as an indicator of user trust.

1.4 SIGNIFICANCE OF STUDY

The subject matter is quite pertinent to contemporary IT problem-solving due to a number of reasons.

- **Digital Accessibility**

This is no longer a choice; it is a legal and ethical obligation in a number of jurisdictions. With the web making the main interface to the government service, education, and banking, the failure to create interfaces that are not conducive to people with disabilities is one of the key failures of the IT infrastructure. This paper offers a technical remedy of protecting inclusivity automatically.

- **The Future of Web Development**

This is Personalized User Experience. Users are now demanding contextual interfaces. This work contributes to the HCI research since it is no longer a field of static responsive design, but is now a field of intelligent design.

- **Trustworthy AI**

This project resolves a critical problem in the current IT ethics of ensuring that humans remain in control of artificial intelligence systems by incorporating a Human-in-the-Loop.

1.5 SCOPE OF STUDY

- **Systematic Scope:** This paper deals with Web User Interfaces (DOM-based environments) and it will not be generalized to native mobile apps and computer software. The specified focus of AI is Constrained Reinforcement Learning 4 in particular, as opposed to other types of machine learning, such as supervised learning.
- **Functional Scope:** The system will be constrained to certain UI adjustments that include font size changes, contrast adjustments, simplifying the layout, and element sizing. It will neither create new content nor paraphrase.
- **Geographical Scope:** The project is executed at a local level, but the standards that are applied in it (WCAG 2.1) are International Standards (ISO/IEC 40500) on a global level. Thus, the solution applies to the international context of the World Wide Web.

1.6 JUSTIFICATION OF STUDY

The solution will add to the literature on Computer Science and Human-Computer Interaction (HCI). Available literature usually considers the concepts of “Accessibility” and “Optimization” as individual objectives. This paper supports its method by showing that they can be coupled together using Constrained Reinforcement Learning. Through its example of how an AI agent can be compelled to be made available through a safety layer, this paper presents a roadmap that can guide the developers of the future to create ethical AI agents. It provides a new architectural style, the so-called WCAG Safety Shield, to which other researchers can apply to ensure that AI does not acquire harmful or discriminatory patterns of behavior.

1.7 LIMITATION OF STUDY

In all academic writings, there are restrictions. For this project:

- **Computational Latency:** The “Safety Shield” requires that checking of accessibility must be carried out in real-time before an action is carried out. This can also create latency, which can be a problem for the system performance, a constraint that optimization might not completely overcome in this research. The agent will be trained and tested mainly in a simulated environment using artificial user personas. As this demonstrates the idea, it might not be that representative of the unpredictable and chaotic nature of human behavior in the real world.
- **User Resiliency of Automated Checks:** The system uses automated devices to check the compliance of WCAG (such as axe-core). It is a well-known weakness that automated tools can only identify the accessibility problems that are around 30-50 percent; some problems are beyond the scope of this automated system and need to be handled by humans.

1.8 DEFINITION OF TERMS

- **Adaptive Web Interface (AWI):** A user interface that automatically adapts its design, content, or structure depending on the context and behavior of the user.
- **Reinforcement Learning (RL):** This form of machine learning is where an agent makes decisions by taking actions in a setting and being rewarded or punished.
- **Constrained Reinforcement Learning (CRL):** This is a type of RL where the agent strives to maximize rewards, but must meet a cost or safety constraints.
- **Human-in-the-Loop (HITL):** A paradigm in which a human is needed to ensure, correct, or direct the actions of an AI system.
- **WCAG (Web Content Accessibility Guidelines):** A set of guidelines that have been issued by the W3C to ensure that web content is more accessible to individuals with disabilities.
- **DOM (Document Object Model):** The computer interface of web documents, the structure of the page in the form of a tree of objects, which are processed by the AI agent.
- **Safety Shield:** This is a software component of this structure that imitates an action and compares it to safety rules (WCAG) before it is allowed to occur.

CHAPTER TWO

LITERATURE REVIEW

2.1 Concept of Web User Interfaces

A web user interface, abbreviated as UI, refers to the interactive part of a web-based application that helps users interact with an application with the aim of achieving particular goals. It is the visible part of a web-based system, which is used by users to view information, input information, navigate between pages/modules, as well as getting feedback from the system. The web UI is made up of web UI elements, which include navigation menus, buttons, hyperlinks, web forms, icons, search boxes, tables, as well as system messages. The effectiveness of a web application is determined by the quality of the web user interface, as users are able to understand the functions of a system using the web UI, which helps them execute their duties quickly.

Web user interfaces are typically implemented using web technologies such as HTML, CSS, and JavaScript. HTML is used for structuring content on the web, while CSS is used for its layout. JavaScript is used in implementing dynamic features like input validation and providing feedback to the users. With the growth of the evolution of web applications, the web interface is now being used to interact with the server through application programming to access the information needed to be showcased to the users. Hence, the web interface is not purely just the display object anymore but is considered to be the dynamic platform where changes can take place due to the dynamic changes of the system as well as the changes caused to the states of the web interface by the users. Research has confirmed the dynamic platforms to be the adaptable platforms, which can also optimize the web interface layouts. A typical web UI consists of a number of different components, some of which play a critical role in enabling user interaction. Among them, the navigation components include menus,

tabs, breadcrumbs, and sidebars, and the form components include forms and inputs. Input components, such as text boxes, drop downs, radio buttons, and file uploads, enable users to supply information or change system settings. Action components, such as buttons and links, enable users to complete actions, such as submitting a form, acknowledging the execution of actions, or transitioning between the stages of a business workflow. Feedback components, such as alerts, notifications, load indicators, or progress bars, let users determine the status of the system or the successful execution of actions. Users become able to work faster with a lower error rate.

Usability stands out in the importance of UI designs for the Web because it determines the ease of learning and use of the interface for the purpose of task completion. Efficiency of use, consistency, effective navigation, and error prevention characterize a usable Web interface. Nevertheless, recent theories related to adaptive interfaces illustrate the insufficiency of performance measures for UI evaluation. Even if the speed and accuracy increase, user satisfaction may decrease if an adaptive system makes predictions that lead to an unpredictable or confusing interface. For example, studies on user interface adaptation are focused on research that highlights how interface satisfaction would need to be observed from the point of view of performance as well as comfort.

Another critical concept of web user interfaces is that of accessibility and inclusive design. Accessibility involves ensuring that people with disabilities, whether visual, motor, or cognitive, can successfully access and use the content of the web by means of assistive technologies. Inclusive design then encompasses accessibility in that it aims to include an even wider range of users in different settings. It appears to be especially important in regard to modern adaptive web interfaces in that dynamic

changes in the UI could inadvertently pose issues of accessibility, such as when navigation order is harmed, readability becomes an issue, or confusion arises. A systematic review of machine learning in the area of adaptive accessible user interfaces illustrates that many adaptive systems do not set accessibility as an absolute criterion, thus forming a research gap in regard to that.

Conclusion In conclusion, the definition of web UI includes the interactive elements for supporting user activities, the techniques for facilitating web interaction, and the concepts of usability and accessibility to support effective system usage. With the further development of web technology, there is a greater need for web UI to shift to intelligent and adaptive interfaces, and, hence, a need to achieve a balance of personalization and system stability, and accessibility.

2.2 Evolution of Web User Interfaces

The development of web user interfaces also mirrors the ever-increasing pace of web technology advancement and the need for greater interactivity from web users. During the initial development of web technology itself, the design of most web user interfaces was static. This means that web pages were designed mostly for the purpose of information presentation. On the user side, only the simplest of functional capabilities were available to the user. These included simple tasks like reading content, following hyperlinks for browsing, and filling out simple forms. This was the initial development of web user interfaces.

As the usage of the internet increased on the web, the interfaces that were used on the web also started to change from static interfaces to dynamic and interactive interfaces known as web interfaces. Such a change in the interfaces was motivated by the need to improve the experience and system response to the activities performed by the users through the click of a mouse or input into the form. Because of the invention of

dynamic web interfaces, users were presented with dynamic web interfaces that reacted instantaneously to activities performed by the user, such as the time taken to click a certain link and input information into the form. Such a change contributed to an improvement in usability due to the improved time taken by the system to react to the activities performed by the user. Web applications currently also offer the capability to update interfaces in real time, a factor that also contributes to the interface design factor as it does not only affect the user interface design but also the way the system responds to the activities performed by the user.

The following significant stage in the evolution of web UI came under the impact of the rise in the use of smartphones, tablets, and multi-device web browsing. The following significant stage in the evolution of web UI came under the impact of the rise in the use of smartphones, tablets, and multi-device web browsing. Users started to use web systems on various screen sizes, along with various input mechanisms such as touch, stylus, and voice. This gave birth to the concept of responsive web design, which allows the layout of a web system to automatically change according to the device used. Although responsive web design is quite helpful in improving the accessibility of the web system with respect to various devices, yet responsive design itself failed to cater to the differences in user preferences, abilities, and accessibility requirements. Therefore, researchers and designers shifted towards adaptive web interfaces.

Adaptive UIs are a major advancement area for web UIs since they act to adapt the UI design depending on user behavior or needs. For instance, an adaptive interface might rearrange menu items according to often-used tasks, improve navigation for new users, or point users toward favorite items. On the other hand, some of the challenges associated with adaptive interfaces might include a lack of predictability and control

by users when changes occur at a very high frequency. Gaspar-Figueiredo et al. (2025) indicate that adaptive interfaces can positively impact performance but can also impact user comfort and satisfaction levels depending on user perceptions of interface changes.

However, when web applications became more complex, scientists added the concept of Artificial Intelligence (AI) or machine learning to improve the quality of adaptation. Originally, adaptational systems were depending upon human rules; nonetheless, these are restrictive since user behavior is dynamic. Reinforcement learning (RL) was a major intelligent approach adopted since it considers interface adaptation as a decision-making task. The system is able to observe the patterns of how users interact with the system, make interface changes, and learn a policy on how to adapt the interfaces based on reward signals. An example of the advancement is showcased in Sun et al. (2024), the adaptive user interface generation using reinforcement learning. Recent research also indicates the potential for issues with trust and usability as a result of completely automated adaptation, unless users are aware of the reasons for the modifications made to the user interface. Hence, human-in-the-loop (HITL) techniques emerged as a result, which include the consideration of user feedback for adaptation processes. Gaspar-Figueiredo et al. (2025) highlight the significance of human feedback and how it could increase the acceptability of reinforcement learning-based adaptive interfaces for users by ensuring that the adaptation process is based on user preferences.

This represents the contemporary era of web user interfaces, which have evolved from completely automated user interfaces to human-centered interfaces for adaptation and are also levelling the playing field for web user interfaces toward user-controlled interfaces for web services and applications.

To this growing list, the trend in web user interfaces concerning matters of accessibility and inclusiveness also needs to be included. The contemporary web UI interface is very interactive, as its design changes dynamically. This actually creates accessibility challenges, as such changing interfaces are a barrier when a user is using an assistive technology or a navigation system that does not change. Hence, the question of inclusiveness arises as an important factor in the design of a web UI. In a nutshell, the historical account of the web UI design has to be depicted as an evolution from a static presentation of information to an intelligent adaptive one, passing through such important milestones as the appearance of responsive and interactive ones in the way they are presented to the user. As web UI has advanced, there have been chances for enhanced personalization; on the contrary, there are also some challenges to be met with.

2.3 Web Accessibility and Inclusive Design

Web accessibility can be defined as the design and development of web systems to ensure that people with disabilities can perceive, understand, navigate, and interact with web content effectively. Web accessibility is crucial because the web has become a very significant platform for learning, communication, trade, and service provision. If the web interface is not made accessible, it can prevent people with disabilities from using web systems fully. Web accessibility can be viewed as inclusive design, which ensures that the web interface is useful for a broad range of people, such as people with permanent, temporary, or situational impairments like poor lighting or low-end internet connections.

Accessibility of web interfaces refers to a number of needs, including those for visual, hearing, physically, speech, and cognitive impairments. For instance, web users with visual impairment may make use of screen readers or need contrast and scalable

text. Users with physical disabilities might use keyboard-only browsing and other alternative interfaces. Users with cognitive disabilities might especially benefit from simple layouts, predictable movement, and streamlined interaction flow. Such needs demonstrate that accessibility not only involves adding assistive functionality, but sometimes involves creating interfaces less burdensome and easier for everyone to use.

An international standard used in making a website accessible is known by the acronym WCAG (Web Content Accessibility Guidelines). This is a standard that focuses on making web content accessible and is built around four principles: "Perceivable, Operable, Understandable, Robust" (POUR). Despite WCAGs availability and guidelines in making a website accessible, studies indicate that current dynamic and interactive web systems are not meeting accessibility standards despite these standards being in place and widely used in making websites accessible.

A lot of web applications have dynamic updates, animations, and interactive elements that, if not carefully planned, may raise accessibility issues. For instance, in cases where changes occur in the content without any notifications, it might become difficult for the screen reader software to recognize changes in the content. Moreover, keyboard focus might become an issue whenever dynamic changes take place in the arrangement of the content, which might not allow people not using the mouse to navigate through the content easily. Such issues could typically arise in modern web applications, which largely depend on interfaces developed through JavaScript.

Recent studies affirm that accessibility solutions are continually improving but that work is left when it comes to integration and consistency. Bercaru and Popescu (2024) analyzed techniques and tools applied in accessibility in online systems ranging from voice interfaces to speech recognition and assistive technologies. Their research

affirmed that despite various tools being developed and readily available, accessibility is yet implemented in standalone aspects and not integrated from start to finish within interface design. This leaves systems looking accessible in some respects but also imperfect in their navigation interface or content organization aspects.

This supports why accessibility is also an aspect that requires inclusion from start to finish in UI design and development.

Inclusive design also emphasizes that the needs of users are very diverse, especially in vulnerable groups like senior citizens. Wickramathilaka et al. present that seniors usually have many interaction barriers because of reduced vision, reduced precision in motor tasks, or because they cannot comprehend complex layouts. Their findings confirm using adaptive and accessible interfaces which can adjust the layout, readability, and navigation according to the user's needs. This indicates that inclusive design is not about mere compliance but flexibility and personalization for the facilitation of diversified users.

Color and contrast are the other important accessibility issues in terms of web interfaces. Many systems use color to represent meaning; for example, red can mean an error that has been made, while green means success. However, such cues may be difficult or poorly perceived by users with color vision deficiency. Sajek et al. (2025) investigated the perception of accessibility of resources on the web for users with color vision deficiency and proved their point on how color and contrast affect the usability and readability of a resource on the web. According to the result, it is essential to make the web system enforce the use of accessible colors and design to make sure that resources on the web do not contain information conveyed by colors.

The relevance of accessibility further escalates in adaptive web interfaces, as the adaptation process may introduce additional accessibility issues. For instance, if the

menu is rearranged in the adaptive system for personal use, the keyboard navigation may become distorted. A systematic review published on machine learning in adaptive accessible interfaces suggested that current adaptive systems based on machine learning do not always consider accessibility as a hard constraint while adapting in real time, making room for accessible adaptive design.

In a nutshell, web accessibility and inclusive design play a vital role in modern web-based user interfaces because they ensure systems can be used effectively by diverse users, including people with disabilities. Although accessibility standards and guidelines give strong guidance, research indicates that many systems face serious problems while integrating accessibility into their interfaces, especially as they grow increasingly dynamic and adaptive. Consequently, adaptive web interface design should take up the challenge of accessibility constraints and inclusive principles to make sure personalization does not reduce usability for any group of users.

2.4 Adaptive Web Interfaces and Intelligent Learning Approaches

Adaptive web interfaces are a type of web user interface that can adapt its content, layout, structure, or interaction behavior to meet the needs of the user. Unlike static web interfaces that are designed for all users in a similar way, adaptive web interfaces are designed to make the user interface experience as personalized as possible to make interaction as easy, quick, and efficient as possible. User adaptation can include adjusting menus to the user's needs or requirements, restructuring learning processes for new users, or adapting visual aspects such as font size or color for better visualization. This has major importance since all users have different skills levels, needs, or devices to work or interact with a computer system.

Confusion arises often when adaptive interfaces are considered to be the same thing as so-called responsive design. Responsive design primarily changes the layout in accordance with the screen size so that it can be used for desktop computers, tablets, and mobile phones. Nevertheless, adaptive interfaces take a further step in adapting the interface according to user-related factors like user behavior patterns, usage data in terms of interaction histories, and accessibility requirements. While adaptive interfaces are more user-centric in their approach, they also face several issues related to interface instability and loss of predictability and user trust when unexpected changes take place.

One of the most critical issues that must be considered within adaptive web interfaces is that of ensuring improvement in usability rather than confusing the user. Several studies demonstrate the enhancement of task performance time and optimization of the adaptive approach, though the user may feel negative in terms of the adaptive environment in the event of frequent changes within the user interface. Gaspar-Figueiredo, et al. in 2025 verify that performance indicators alone must not be used for adaptive interfaces, due to the significant role of user preference in the acceptance of adaptive systems.

First, adaptive interfaces in the early days were rule-based, and the designer would set up conditions such as “if a user is new, display simpler features” or “if a feature is heavily used, move it up.” Though rule-based adaptation can be easily implemented, it has limitations in terms of its ability to adapt to complex patterns of user behavior or changes in the needs of the user over a period of time. As web applications became complex, intelligent learning techniques, namely machine learning, were used to adapt systems. There is a systematic review of machine learning for adaptive

accessible interfaces available that recommends the use of learning for adaptation but states that accessibility is still not a requirement for adaptive systems.

Reinforcement learning (RL) has turned out to be a most effective learning method for adaptive web interface design as it treats user interface adaptation as a sequential decision-making task. Here, the reinforcement learning agent takes into account the state of the user interface as well as the user environment. Then it makes a decision (like a different layout or menu arrangement) and receives a reward as feedback. Sun et al. (2024) have shown that reinforcement learning can be used to adapt the generation of the user interface by learning that the layout of the interface can be adapted to be optimal. It has important applications in that it is independent of hard-coded rules.

Nevertheless, reward-based adaptation in reinforcement learning is greatly dependent on reward functions. For example, if reward functions are centered only on metrics of engagement activities like clicks and time used, adaptation systems may end up learning and implementing adaptations that might enhance user engagement but compromise user usability and accessibility. Vidmanov (2024) discusses that modeling rewards for usability is a key area that can help make sure that improvements are done by reinforcement learning without implementing negative interface adaptations by design.

Another key development in the area of adaptive web interfaces is the incorporation of human feedback into intelligent learning systems. With human-in-the-loop (HITL) adaptation, users are able to shape and inform interface adaptation processes and thereby increase the levels of transparency and trustworthiness associated with them. According to Gaspar-Figueiredo et al. (2025), the incorporation of human feedback into reinforcement learning algorithms helps enhance the acceptability of adaptive

modifications and align the modifications with user expectations. Similarly, the significance of human-in-the-loop reinforcement learning, which improves safety, trust, and alignment, particularly in automated decision-making impacting human experience directly, is emphasized in the study done by Retzlaff et al. (2024). This is crucial for an adaptable web interface because, sometimes, users might not want systems that adapt unpredictably without control or explanations.

Context-aware adaptation is another intelligent method that further refines adaptive interfaces. Context-aware interfaces are based on adapting decisions influenced by the context, which might include device information, location awareness, time of interaction, and user goals and interaction history. Adaptation based on context was also discussed by Carrera-Rivera et al., (2024) in the context of proposing the AdaptUI system, which is a context-aware system that enables model and recommendation-based adaptation. The paper clearly demonstrates that the modeling of context enables the suppression of random adaptation and the relevance of personalized interfaces. The limitations of accessibility are still required for better integration for adaptation for every user.

In conclusion, adaptive Web interfaces are a significant development in web interface design as they have made it possible to adapt to user needs. Although reinforcement learning has immense potential for adaptation and learning in a continuous manner, yet its application needs a well-defined reward function to avert usability or accessibility issues through correct handling of constraints. Human-in-the-loop and context-driven learning strategies further advance adaptive systems to boost trust, transparency, and relevance through intelligent learning methods that form a robust foundation for building adaptive web interfaces.

2.5 Related Works

This section presents an extensive discussion based on sixteen current research articles that have appeared in journals and conferences between 2020 and 2025 and are highly related to adaptive web interfaces, reinforcement learning-based adaptive user interface systems, usability evaluation, accessibility, and inclusive design issues. Each of the research articles will be considered separately in this section with an emphasis to identify their contribution to the field to understand their importance and to understand the gap that leads to the need for an accessibility-conscious adaptive web interface model with well-controlled adaptation mechanisms.

A rather encompassing systematic literature study on the application of machine learning technologies in adaptive and accessible user interfaces was carried out by Kristić et al. (2025). A large scope of adaptive interface technologies developed in conjunction with supervised, reinforcement, and even hybrid machine learning approaches was surveyed in this study. From the study, the following were established: While the application of the aforementioned advanced technologies is rather beneficial in terms of improving the overall personalization capabilities of the adaptive interface, the inclusion of accessibility attributes can be said to be optional or subordinate. For instance, the study found that most adaptive technologies were primarily designed with the intention of optimizing the entire performance as well as the level of engagement that users can enjoy with the adaptive interface. Moreover, the inclusion of the accessibility attributes, such as WCAG, was rather limited or even lacking altogether in the adaptive technologies surveyed. This study is rather relevant with regard to the present study since the gap relating to the inclusion of the aforementioned advanced technologies to adaptive user interfaces is acknowledged.

Authors Sun et al. (2024) have proposed an adaptive user interface generation process based on reinforcement learning techniques. According to Sun and his colleagues, personalization can be defined as a process of sequential decision-making. In their study, Sun and his colleagues have represented the interface adaptation process as an interactive system environment that enables an agent to learn and eventually gain an optimal interface configuration based on observations and feedback received based on a reward signal. According to Sun and his colleagues, their study revealed that reinforcement learning techniques can generate interface configurations and optimize them without heavily relying on personalization techniques or interface generation algorithms. Nevertheless, as reported by Sun and his colleagues, personalization heavily depends on an adequate reward signal; otherwise, it might emphasize user engagement rather than actual usability. Moreover, personalization constraints regarding interface accessibility were not carefully emphasized in this particular study. This particular study relates to the present research as it established that reinforcement learning techniques are applicable to adaptive interface generation techniques and emphasized that personalization performance heavily depends on adequate control and constraints regarding interface accessibility.

Gaspar-Figueiredo et al. (2025) sought to assess the usability of adaptive user interfaces by examining the differences in the performance and preferences of users with regard to non-adaptive and adaptive menus. The researchers carried out experiments and assessments using the evaluation measure. The results of the research revealed that the use of an adaptive interface can improve the efficiency and accuracy of performing tasks. However, the research confirmed that users are likely to specify a nonadaptive interface. The researchers argued that the assessment of the efficiency of an adaptive interface cannot rely only on the results obtained from the experiments

and assessment. Other factors, including the satisfaction level of the users, the level of trust with the interface, and the level of control by the users, are crucial. These factors are very critical in a design evaluation. The research supports the current research by covering topics that are crucial and crucial in the design of web interfaces.

In a similar research, Gaspar-Figueiredo et al. (2025) developed a new reinforcement learning-based approach to building an adaptive interface and proposed a new adaptive interface framework that utilizes the element of humans. In the research, the authors tackled the concept of building an adaptive web interface that included mechanisms that could utilize the element of humans. The results demonstrated that the incorporation of the element of humans into the adaptive interface provides a platform where the user can either accept or reject the altered interface. The research concluded that the full autonomy of building an adaptive system could result in a high level of frustration among the users. Such a research is highly applicable and significant to the current research as it provides the required evidence that justifies the involvement of the element of humans.

Vidmanov (2024) investigated the usability reward modeling technique employed in reinforcement learning-based adaptive user interfaces. In the paper, the researcher examined the effects of different reward models on the usability of the computer interface. In the research, the results showed that several reward models focusing on efficiency, reduction of errors, and satisfaction of the user can offer more predictable interface adaptations compared to those focusing on the engagement of the user. The researcher highlighted the fact that inappropriate reward models can lead to inconsistent interface changes. The paper is relevant to the research objectives since

the researcher highlighted the significance of reward models to ensure the usability of the web interface.

Similarly, in search of the potential of deep reinforcement learning methods for improving the user interface in the context of the user interface system design framework provided in this paper, a paper titled "Deep reinforcement learning methods for enhancing the user interface in the context of the user interface system design framework" was published by Khamaj in 2024. Its relevance to the topic of the presented paper emerged from the demonstration of the potential of learning-based user interface system design on the one hand; on the other hand, the author of the paper warned the user interface system design framework against the potential pitfalls of unrestrained learning in the context of user interface system design for a better user interface system within user interface systems. Specifically, the potential pitfalls mentioned in the paper published by Khamaj in 2024 revolved around the possibility of unstable user interface system design in the context of the user interface system design framework provided in this paper.

Carrera-Rivera et al (2024) developed a framework that was termed "AdaptUI." The newly introduced framework was greatly motivated by the need to "facilitate adaptive user interface development." The newly introduced framework greatly underscored the need to "develop a comprehensive model of user and system context, including user context—user preferences, user behavior, user environment, as well as a corresponding notion of system context, such as interaction devices' capabilities, interaction conditions, etc." The newly introduced framework was analyzed through various case study applications, proving its ability to "show that user satisfaction is greatly influenced by the ability to adequately reason about user context changes

during user interaction.” The results revealed that its ability to “support users interacting with a user interface that can reason about user context changes is critical to the overall success.” The problem attempted to tackle functional personalization, with issues regarding accessibility only being implicitly tackled. The problem is very relevant to the research because, through its findings, “the importance of user context is revealed when analyzing user satisfaction—a critical aspect to be incorporated into adaptive user interface development.”

Bercaru and Popescu (2024) in their study created an in-depth survey of accessibility approaches and difficulties in web interfaces focusing on the various applications of artificial intelligence in accessibility enhancement. Some of the approaches discussed in the study were recognition systems, auto-capturing systems, optimizing screen reader systems, and intelligent assistive technologies all geared towards enhancing the accessibility of web interfaces for individuals living with various impairments. Despite the promising features of the discussed approaches in enhancing accessibility in web interfaces, the study established the main focus of all the approaches as being supplementary features rather than integral features in interface development. Another key aspect established in the study was the late inclusion of accessibility in interface systems as an auxiliary aspect of interface development. This aspect of diversity in interface approaches presents an implication of the negative perspectives of the current accessibility approaches in the context of promoting interface accessibility. This study strongly enhances the views of the current study in the context of the negative perspectives of the current accessibility approaches in the context of promoting interface accessibility.

The research paper by Sajek et al. (2025) explores and presents the views and perceptions related to web accessibility among users who face color vision deficiency. The discussion in the study has emphasized the use of improper use of contrasts and color combinations. Similarly, the research has explained how the use of colors can negatively affect the readability of online content. In the research study, the author has clearly explained how color-related elements can be somehow inappropriate and cannot be an effective option when other visual options are not available. The primary focus of the study was on showing how modern website designs overlook accessibility requirements and thus violate certain standards. The research has emphasized and explained that adaptive website designs, where certain operations and functions change frequently, can negatively affect accessibility. According to the research authors, adaptation is one of the primary features of website designs and thus can somehow violate accessibility requirements. The study has sufficiently explained how and why adaptation can be incorrect and has proposed certain requirements to be included in the adaptation process of website interfaces, and thus the research is relevant and sufficient.

Wickramathilaka and Müller (2025) presented a model-based approach that is applicable to the design and development of adaptive and accessible user interfaces especially for seniors. Their work was a response to the rising usability challenges that affect the elderly population due to cognitive, visual, and physical impairments that come with old age. From their study, the model used adaptive layout through the dynamic modification of the interface to improve the experience of the user. More especially, the adaptive model was effective in the improvement of the reading experience for the people sampled from the experimental design. The findings indicated that the model was effective when compared to the ones that are static.

Access to the user interface is critical and needs to be adapted to the needs of the seniors, crucial since the accessibility needs are highly diverse compared to the averages that are one-size-fits-all, making accessibility adaptation crucial since it needs to be personalized to the needs of the individual.

Jamali et al. (2025) investigated AI-powered adaptive learning interfaces for evaluating the effect of personalization on user engagement and motivation. The paper approached this by quantitatively and qualitatively measuring how various adaptive strategies influence learning outcomes and user perception. Results have shown that well-designed personalization significantly enhances user satisfaction, engagement, and performance, whereas poorly implemented adaptation strategies lead to confusion, frustration, and reduced acceptance. Transparency and predictability of adaptive behavior have been pointed out, with critical remarks that users may respond more positively when adaptation goals are clearly aligned with their expectations. Adaptation should, therefore, be goal-directed and carefully tuned to user needs. These insights are particularly relevant for adaptive web interface design, as they further point out the necessity of controlled, user-centered adaptation strategies that do not introduce arbitrary or excessive changes.

Retzlaff et al. (2024) conducted a comprehensive survey on human-in-the-loop reinforcement learning approaches for adaptive systems. The present study reviewed how the integration of human feedback into learning processes enhances system reliability, safety, and interpretability. Results showed that human oversight greatly enhanced users' trust in the system and further reduced the risks involved with autonomous decision-making. The authors noted that the provided mechanisms of human feedback allow an adaptive system to correct undesirable behaviors while

learning. This is of particular importance for domains in which errors might negatively affect users. The study also pointed to the importance of human-in-the-loop approaches with respect to accessibility-sensitive applications. These findings strongly support the development of adaptive web interfaces with integrated human supervision, such as the one proposed within the present study.

Jiang et al. (2025) discussed the application of a deep Q-learning approach in the development of the adaptive user interface framework targeting wearable medical devices. While the usability, task effectiveness, and overall user engagement were enhanced with the application of adaptive user interface strategies, there were obvious safety implications with the unbridled application of adaptive user interfaces, as emphasized Jiang et al. This study indicated the need for the application of a constrained learning approach while targeting the adaptive web interface development process. This emphasis on the application of constrained adaptive development is the same as the emphasis that the present research emphasizes on the application of constraints in web usability development.

Troussas (2025), in a study on reinforcement learning-based adaptive interface models using fuzzy dynamic weighting to handle uncertainty in user models, showed that hybrid intelligent methods can cope more efficiently with user preference and usage variability. Thus, a more flexible and robust adaptability using both reinforcement learning and fuzzy logic can lead to more stable models of intelligent adaptability, especially in complex environments where user models are unpredictable in practice. Thus, this study can add to generalizations in intelligent adaptability as a tool in designing more intelligent methods of adaptability in web interface designs using sophisticated intelligent learning methods

In particular, Wang et al. (2024) investigated an issue of effective user interfaces while developing software concerning the management of chronic health conditions. The findings showed that the usability of software, as well as increased adherence among users, were greatly enhanced by the use of an adaptive approach. The authors' emphasis on effective and exhaustive methodology, however, raises concerns about ensuring an effective approach. The issue of stability is again raised, thus providing further backing to the issue of controlled approaches, making way for the objectives of the present study.

Ali and Chen (2024) developed a conceptual framework for self-adaptive mobile user interfaces, focusing on structured decision-making processes for adaptation. The key components that would anchor adaptive behavior were outlined as monitoring, analysis, planning, and execution. Even as the core of this study rested on mobile interfaces, the bases of this principle can be applied to web-based environments. The authors have identified the necessity for adaptation driven by usability and mechanisms for systematic control. This provides the grounding for the present study, reinforcing the application of structured and well-governed adaptation strategies in adaptive web interface design.

2.6 Summary of Related Works

The literature reviewed (2020-2025) suggests adaptive interfaces for the Web can be used for enhanced usability/personalization by use of smart approaches like reinforcement learning, but there are issues in the enforcement of accessibility, reward schemes, stability, and trust for the users.

Author(s) & Year	Title of Article	Focus	Key Finding	Gap Identified
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Kristić et al. (2025)	Machine Learning for Adaptive Accessible User Interfaces: A Systematic Literature Review	Review of ML techniques in adaptive and accessible UIs	Demonstrated that supervised, reinforcement, and hybrid ML techniques significantly enhance personalization and interaction efficiency. However, accessibility considerations such as WCAG compliance were often treated as optional extensions rather than core design objectives. Most systems optimized engagement and performance rather than inclusive usability.	Accessibility constraints are weakly enforced or absent during runtime adaptation, indicating a lack of accessibility-aware learning frameworks.
Sun et al. (2024)	Reinforcement Learning-Based Adaptive User Interface Generation	RL-driven UI personalization	Modeled UI adaptation as a sequential decision-making process where agents learn optimal configurations from user interaction feedback. Showed that RL can autonomously generate effective interface	Accessibility and usability constraints were not embedded in the reward function, leading to a risk of engagement-driven rather than usable or accessible adaptations.

			layouts without predefined personalization rules.	
Gaspar-Figueiredo et al. (2025)	User Experience with Adaptive User Interfaces	Usability evaluation of adaptive menus	Found that adaptive menus improve task efficiency and accuracy compared to static menus. However, many users preferred non-adaptive interfaces due to trust, predictability, and control concerns.	Usability evaluation often ignores user trust, transparency, and accessibility when assessing adaptive systems.
Gaspar-Figueiredo et al. (2025)	Human Feedback in Reinforcement Learning for Adaptive User Interfaces	Human-in-the-loop RL for UI adaptation	Demonstrated that incorporating human approval or rejection of adaptations significantly improves user satisfaction and reduces frustration caused by fully autonomous systems.	Accessibility constraints were not explicitly modeled in the human feedback loop.
Vidmanov (2024)	Usability Reward Modeling in Reinforcement Learning-Based Adaptive Interfaces	Reward modeling for RL-based UI adaptation	Showed that reward functions emphasizing efficiency, error reduction, and user satisfaction result in more stable and predictable adaptations	Accessibility standards and human oversight were not considered in reward formulation.

			than engagement-focused rewards.	
Khamaj (2024)	Deep Reinforcement Learning for Enhancing User Experience in Adaptive Interfaces	Deep RL in adaptive UI systems	Confirmed that deep RL agents can effectively personalize interfaces by learning user interaction patterns, improving usability and performance. Warned against instability from unconstrained learning.	Lack of explicit control, accessibility constraints, and safety mechanisms.
Carrera-Rivera et al. (2024)	AdaptUI: A Framework for Adaptive User Interfaces in Smart Product–Service Systems	Context-aware adaptive UI framework	Established that comprehensive modeling of user and system context significantly improves relevance and user satisfaction in adaptive systems. Demonstrated effectiveness through multiple case studies.	Accessibility considerations were implicit and not formally modeled as constraints.
Bercaru & Popescu (2024)	Accessibility in Online Platforms: Techniques, Challenges and AI-Based Solutions	Survey of AI-based accessibility solutions	Identified speech recognition, captioning, screen reader optimization, and assistive technologies as effective tools.	Accessibility is treated as a supplementary feature rather than embedded across the design lifecycle.

			Highlighted that accessibility is often added late in development.	
Sajek et al. (2025)	User Perceptions of Web Accessibility for Colour Vision Deficiency	Accessibility perception study	Revealed that improper contrast usage and reliance on color cues significantly reduce readability and usability. Highlighted that dynamic visual adaptation can worsen accessibility if uncontrolled.	No intelligent or adaptive accessibility-aware solution was proposed.
Wickramathilaka & Mueller (2025)	Adaptive and Accessible User Interfaces for Seniors	Adaptive UI design for elderly users	Demonstrated that adaptive layouts significantly improve readability and navigation for seniors compared to static interfaces. Emphasized the need for personalized accessibility.	Lack of scalable, automated adaptive accessibility models.
Jamali et al. (2025)	AI-Powered Adaptive Learning Interfaces	Personalization in learning interfaces	Showed that well-designed personalization improves engagement, motivation, and performance, while poorly designed adaptation frustrates users.	Limited focus on accessibility and constraint-based adaptation.

Retzlaff et al. (2024)	Human-in-the-Loop Reinforcement Learning: A Survey	HITL RL methods	Found that integrating human feedback improves trust, transparency, and safety in adaptive systems. Highlighted importance for sensitive domains.	Few applications specifically target adaptive user interfaces with accessibility concerns.
Jiang et al. (2025)	Adaptive User Interfaces for Wearable Medical Devices Using Deep Q-Learning	Adaptive UI in healthcare	Demonstrated improved usability and engagement but emphasized safety risks of unconstrained adaptation in medical contexts.	Necessity for constrained and accessibility-aware learning not fully addressed.
Troussas (2025)	Reinforcement Learning-Based Adaptive Interfaces with Fuzzy Dynamic Weighting	Hybrid intelligent adaptation	Showed that combining RL with fuzzy logic improves stability and handles uncertainty in user behavior more effectively.	Accessibility constraints and human oversight were not integrated.
Wang et al. (2024)	Adaptive User Interfaces in Healthcare Software	Adaptive UI for chronic condition management	Found improved usability and user adherence but stressed the importance of system stability and inclusive evaluation.	Limited guidance on enforcing accessibility and stability constraints.
Ali & Chen (2024)	A Conceptual Framework for Self-Adaptive Mobile User Interfaces	Structured adaptation decision-making	Proposed a MAPE-based framework emphasizing usability-driven	No explicit accessibility-aware adaptation model was defined.

			adaptation and control mechanisms applicable beyond mobile systems.	
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CHAPTER THREE

SYSTEM DESIGN AND METHODOLOGY

3.0 Overview of the System

This chapter describes system design and methodology developed for an adaptive web interface using CRL, along with HITL mechanisms in order to ensure accessibility compliance and improve user trust. The methodology describes the system architecture, data collection process, learning framework, constraint enforcement strategy, and evaluation methods used in assessing system performance.

It proposes a system that dynamically modifies user interface elements through patterns of user interactions but also addresses the WCAG requirements for accessibility. It includes human supervision of adapted decisions to review and make changes to the adaptive decisions for better usability and to increase user confidence.

3.1 System Analysis and Design Strategy

The approaches adopted included scalability, maintainability, and clarity; hence, an appropriate approach was the adoption of modular design. The system is divided into the following major components:

- User Interface Layer
- Interaction Monitoring Module
- Constrained Reinforcement Learning Engine

- Human-in-the-loop feedback module
- Accessibility Compliance Validator

Each module executes a specific function while communicating with other modules for adaptive and accessible user interaction.

3.2 System Architecture

Such a system would have a layered architecture wherein user interfaces through the web interface would be monitored. User behaviors would be fed to the reinforcement learning agent to calculate the adaptations optimally while keeping the accessibility constraints in check. They may or may not be sent to the user for evaluation.

Architecture Description

- Input: Interactions – click, navigation paths and dwell time
- Processing: Policy learning using Constrained Reinforcement Learning with accessibility constraints
- Control: Validation and feedback through "human in the loop" mechanisms
- Output: Adaptation of web interface compatible with WCAG guidelines

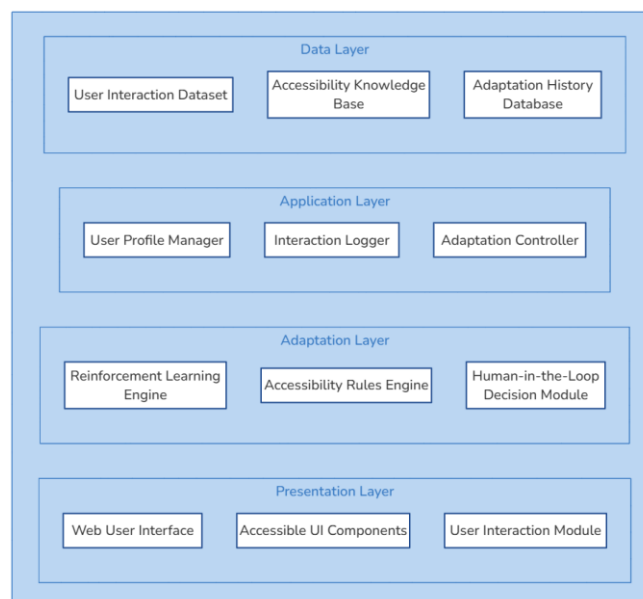


Figure 3.1: Use Case Diagram of the Proposed Adaptive Web Interface System

3.3 Unified Modeling Language (UML) Diagrams

3.3.1 Use Case Diagram

The interaction of system actors with adaptive web interface system can be understood from the use case diagram below:

Actors:

End User

System Administrator (Human Supervisor)

The end user ultimately accesses the web interface, uses elements of the interface, and receives updates to the adaptable interfaces. The system administrator manages the adaptation process by viewing adaptable decisions, approving or editing adaptations, and monitoring accessibility compliance to adhere to WCAG norms.

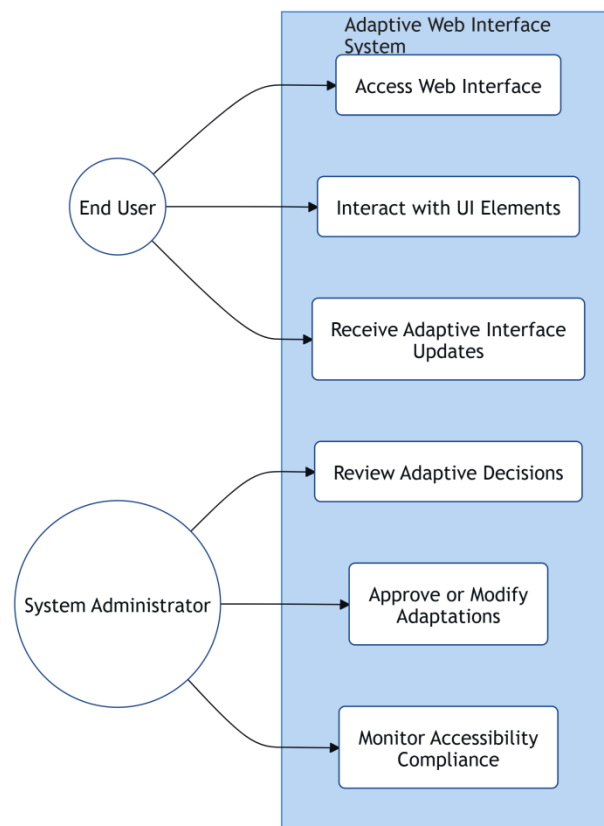


Figure 3.3.1 shows the use case diagram of the proposed adaptive web interface system.

3.3.2 Class Diagram

The class diagram shows the system structure and the necessary components that are part of this system. The necessary components of this class diagram system include: User, Interface State, RL Agent, Constraint Manager, Accessibility Validator, Human Feedback Module.

Each such class has values related to adaptable decisions. Each such class has variables related to accessibility enforcement and human evaluation. These classes have relationships among them showing how user behaviors influence interface states, how the reinforcement learning agent provides adaptation suggestions, how accessibility constraints are enforced, and how human evaluation influences adaptation decisions.

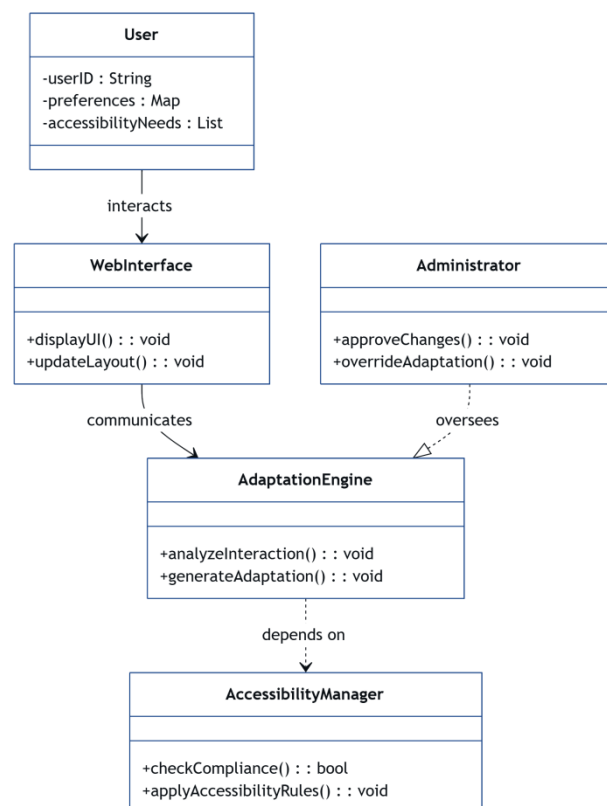


Figure 3.3.2 presents the class diagram of the proposed system.

3.3.3 System Flowchart

The system flowchart illustrates the working of the operational workflow of the adaptive web interface. It starts by considering the interaction initiated from the user's end with the interface. This interaction is captured and analyzed, including the evaluation of the reinforcement learning agent on the current interface state. Further, there are the accessibility constraints that keep the adaptations away from being unsafe or non-compliant. Optionally, human feedback can also be integrated in order to deploy the adapted interface.

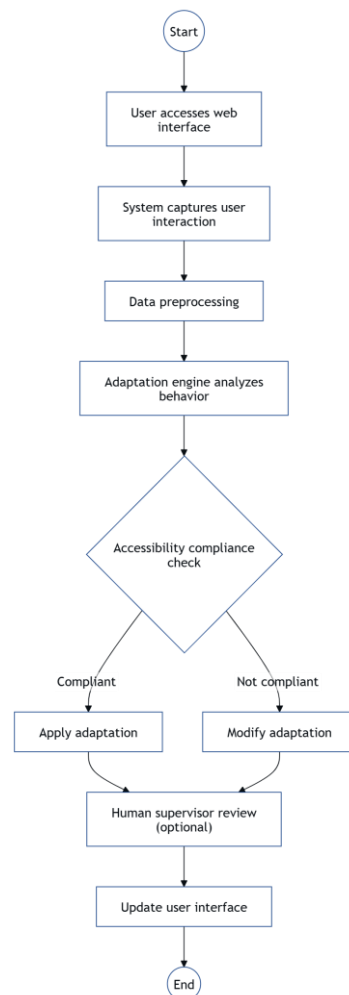


Figure 3.3.3: Flowchart of the Adaptive Web Interface Operation

3.4 Data Collection

User interaction data is captured in a real-time fashion via client-side logging entities integrated with the Web interface. The data captured includes:

- Click frequency
- Navigation sequences
- Session duration
- Interaction errors
- Accessibility-related events (e.g., keyboard navigation use)

The above mentioned data points will reveal much about how users use the system and how it can be utilized well.

3.5 Data Preprocessing

Collected data is processed in some way to enhance data quality. It involves:

- Deletion of incomplete interaction logs
- Normalization of Interaction Metrics
- Aggregation of session-level data
- Encoding of interface states for learning

This preprocessed data is stored in an appropriate structure for the purpose of reinforcement learning.

3.6 Reinforcement Learning Framework

The reinforcement learning framework models interface adaptation as a problem of sequential decision-making, where an agent learns an optimal adaptation strategy through interaction with the environment.

Figure 3.6 is a framework synthesized by the author based on previous studies of reinforcement learning-based adaptive interfaces, web accessibility standards/content

characteristics, and human-in-the-loop systems. Studies included in this synthesis include Sun et al. (2024), Gaspar-Figueiredo et al. (2025), and Retzlaff et al. (2024). It is a framework developed by the author, intended to incorporate these ideas into a cohesive model.

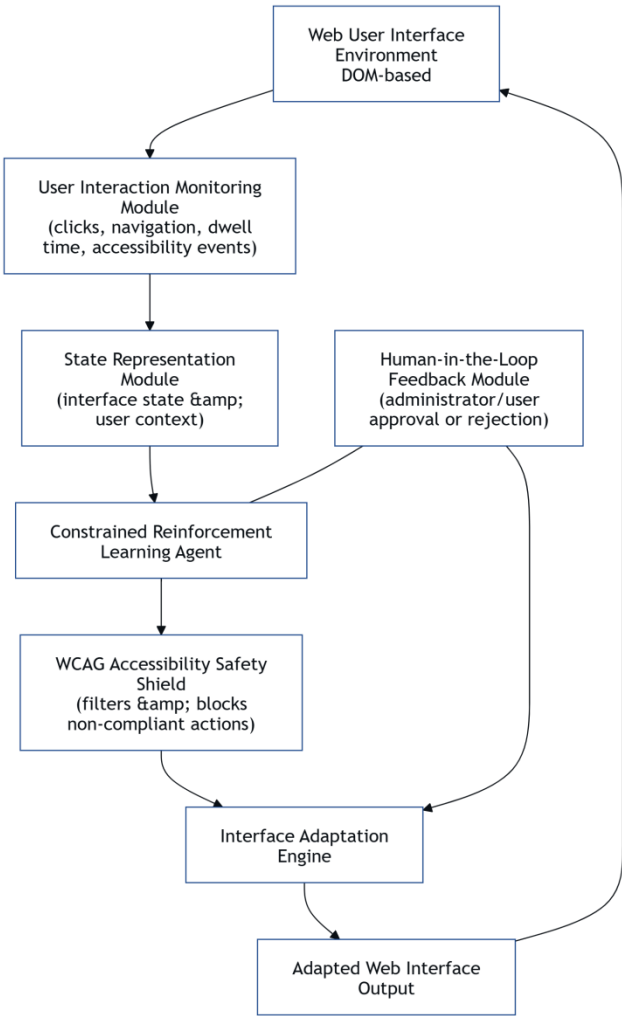


Figure 3.6: Proposed Constrained Reinforcement Learning and Human-in-the-Loop Framework for Adaptive Web Interfaces (Synthesized from Literature and Designed by the Researcher)

3.6.1 State Space

The state space comprises the current configuration of the interface settings and context of the user's interactions, including layout structure, visual settings, navigation complexity, and recent patterns in users' behaviors.

3.6.2 Action Space

The action space consists of permissible interface adaptations, including:

- Font size adjustment
- Contrast modification
- Layout restructuring
- Navigation simplification

Actions that violate accessibility requirements are restricted.

3.6.3 Reward Function

The reward function assesses the effectiveness of interface adaptations depending on:

- Improved task completion rates
- Reduced interaction errors
- Positive user feedback
- Accessibility Compliance

Rewards guide the agent towards inclusive and useable interface configurations.

3.7 Constrained Reinforcement Learning Approach

The accessibility constraints based upon the guidelines of WCAG 2.1 are incorporated in the learning process. The learning process will impose restrictions upon the agent to avoid any actions that will compromise accessibility in aspects such as insufficient contrast between colors, inaccessible navigation systems, or non-accessible text sizes.

The process of constrained learning guarantees that all the learned policies stay safe, inclusive, and compliant.

3.8 Human-in-the-Loop Integration

Human-in-the-loop mechanisms allow users or administrators to:

- Review adaptive decisions
- Provide corrective feedback
- Approve or reject interface changes

Human feedback is incorporated into policy updates, enhancing system transparency and preventing undesirable adaptations.

3.9 Study Population

The target population comprises internet users, who have varying levels of digital literacy as well as differences in their accessibility needs. It will include users with disabilities, like the visually impaired, and those without disabilities to provide a balanced approach.

3.10 Data Analysis Technique

- Both qualitative and quantitative analysis techniques are employed.
- Quantitative analysis of interaction metrics
- Qualitative analysis of user feedback
- Analysis of comparative performance of the interface before and after adaptation

3.11 Training and Testing Strategy

The data set is split into an 80:20 division for training and testing subsets, respectively. Then, the agent of reinforcement learning is trained in an iterative way, while its performance is assessed on data of interactions that have not been seen to ensure its generalization.

3.12 Evaluation Metrics

System performance is assessed using the following parameters:

- Task completion time
- Error Rate Reduction
- Accessibility compliance score
- User satisfaction rating

3.13 Ethical Considerations

Adaptation acceptance rate User privacy and data security are fully incorporated into this interaction type. Anonymity is ensured for all interaction data collected. The interaction is also completely voluntary and adaptive decisions are also fully transparent.

3.14 Chapter Summary

This chapter has covered the system design and its methodology in the development of an accessibility awareness-based adaptive web interface through the use of constrained reinforcement learning. In the next chapter, the system implementation, the achieved results, and the discussion will be presented.

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